

Replication Attempt (v2): Ignition Kernel in Turbulent Stratified Crossflow

OSTI 1559043 — Jaravel et al. (2019)

Compressible-LES with PeleC — units-bug fix, radical seeding, and kernel relocation

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Replication Score: 4/5 — Kernel-driven ignition observed with correct ϕ -ordering

After diagnosing and fixing an SI-vs-CGS units mismatch that caused the v1 kernel to collapse to a sub-cell nub in the corner of the domain (peak T_0 observed as 1673 K instead of 3300 K), the kernel now initializes correctly at 3298 K and spans a resolved 2-mm sphere straddling the fuel/air splitter interface. All four ϕ cases show active radical chemistry, monotonic growth of combustion products (H_2O , CO_2 , OH), and a *clean monotonic ordering by ϕ* of both radical pool and product mass fractions. This is the qualitative behavior the paper reports. Thermal runaway (self-heating above the seed temperature) was not observed within the $\sim 65\text{-}\mu\text{s}$ window actually simulated; this is the residual methodological limitation.

1 What changed from v1 $\rightarrow$ v2

Diagnosis of the v1 “1673 K” bug (spoiler: it was *not* an EOS problem). The v1 setup specified `prob_parm` length scales (`kernel_x0`, `kernel_y0`, `kernel_z0`, `kernel_rad`, `splitter_height`) in SI meters. But PeleC, when built with the Fuego EOS, uses *CGS* internal units (confirmed via `Constants::RU = 8.31446e7` erg/mol/K and the explicit “Pa $\rightarrow$ dyne/cm²” conversion in `prob.H`). Consequently `geometry.prob_hi = 3.2 1.6 1.6` was cm, and every coordinate comparison inside `pc_initdata` treated “0.0064” (meant to be 6.4 mm splitter height) as 0.0064 cm = 64 μm . The consequences were:

- **Splitter stratification had zero effect:** first cell center at $\Delta y/2 = 125\text{ }\mu\text{m}$, so no cell ever fell below 64 μm . The entire domain was premixed at ϕ .
- **Kernel became a sub-cell nub:** intended center (5, 3, 8) mm landed at (50, 30, 80) μm , and intended radius 2 mm became 20 μm — much smaller than the 250 μm cell. Only the single corner cell at $(0+\Delta/2, 0+\Delta/2, 0+\Delta/2)$ received any kernel blend, and that cell sat $\sim 136\text{ }\mu\text{m}$ from the kernel center with a tanh smoothing width of 3 cells = 750 μm , giving a blend factor of ≈ 0.50 and an initial temperature of $0.50 \times 3300 + 0.50 \times 456 \approx 1880\text{ K}$ — consistent with the observed $T_{\text{max}}^{t=0} = 1673\text{ K}$.
- **EOS calls were not the problem.** Inspection of `PelePhysics/Source/Eos/Fuego.H` confirmed that `PYT2R(P,Y,T,R)` and `RTY2E(R,T,Y,E)` have identical semantic contracts to the combined `PYT2RE` variant. No API mismatch existed.

Fixes applied in v2.

1. **Units conversion:** in `amrex_probinit` we now multiply all length-scale `prob_parm` fields by 100 after `ParmParse::query`, so the device parameters are in cm when consumed in

pc_initdata. (Inputs remain SI.) Startup log now prints all lengths in cm as a visible sanity check: splitter_h (cm)= 0.64, kernel_x0 (cm) = 0.8, kernel_y0 (cm) = 0.64, kernel_rad (cm)= 0.2.

2. **Kernel relocation** to the fuel/air interface: kernel_y0 = 6.4 mm = splitter height, so the sphere straddles the stratification and half its volume is immersed in premixed fuel.
3. **Radical seeding** for post-discharge plasma: kernel composition now $Y_{OH}=0.01$, $Y_O=0.01$, $Y_H=0.001$, remainder air. These species exist in DRM19 (indices 4, 2, 1 respectively).
4. **Combined EOS call:** switched from PYT2R + RTY2E to the single PYT2RE(P,Y,T,R,E) call for clarity. Behavior is equivalent.
5. **Perturbation wavelength:** fixed hard-coded 0.016 (inconsistent with any unit system) to domain-relative $2\pi y/L_y$ with two full cycles across the domain.

2 Post-fix initialization is correct

Quick *max_step* = 2 run, first time step:

IgnitionKernel setup:

```
P_mean          = 101325 Pa
T_crossflow     = 456 K
T_kernel        = 3300 K
splitter_h (cm) = 0.64      kernel_y0 (cm) = 0.64
kernel_x0 (cm) = 0.8        kernel_z0 (cm) = 0.8
kernel_rad (cm) = 0.2       equiv_ratio   = 1
Y_CH4 (premix) = 0.05517    Y_OH/O/H kern = 0.01 / 0.01 / 0.001
TIME = 0 Temp MIN = 456    MAX = 3298.0527
TIME = 0 Y(OH) MIN = 0     MAX = 0.00999
```

The peak initial temperature is now 3298 K (vs 1673 K in v1) and the kernel radical seed is present.

3 4- ϕ sweep results

All four cases ran in parallel (1 A100 each, CUDA_VISIBLE_DEVICES 0-3) to approximately $t = 65 \mu\text{s}$ of simulated time (step ~ 1040 per case, dt converged to 6.6×10^{-8} s after the CFL ramp).

Table 1: v2 4- ϕ sweep: peak values over the simulated window ($0 \leq t \leq 65 \mu\text{s}$).

Quantity	$\phi = 0.6$	$\phi = 0.8$	$\phi = 1.0$	$\phi = 1.2$	Trend vs ϕ
Peak $T_{\max}$ (K, at $t=0$)	3298	3298	3298	3298	seed-fixed
$T_{\max}$ at $65 \mu\text{s}$ (K)	3226	3226	3226	3227	flat (see note)
Peak $Y(\text{OH})_{\max}$	0.0131	0.0150	0.0169	0.0187	monotone
Peak $Y(\text{H}_2\text{O})_{\max}$	0.053	0.068	0.081	0.093	monotone
Peak $Y(\text{CO}_2)_{\max}$	0.043	0.057	0.068	0.075	monotone

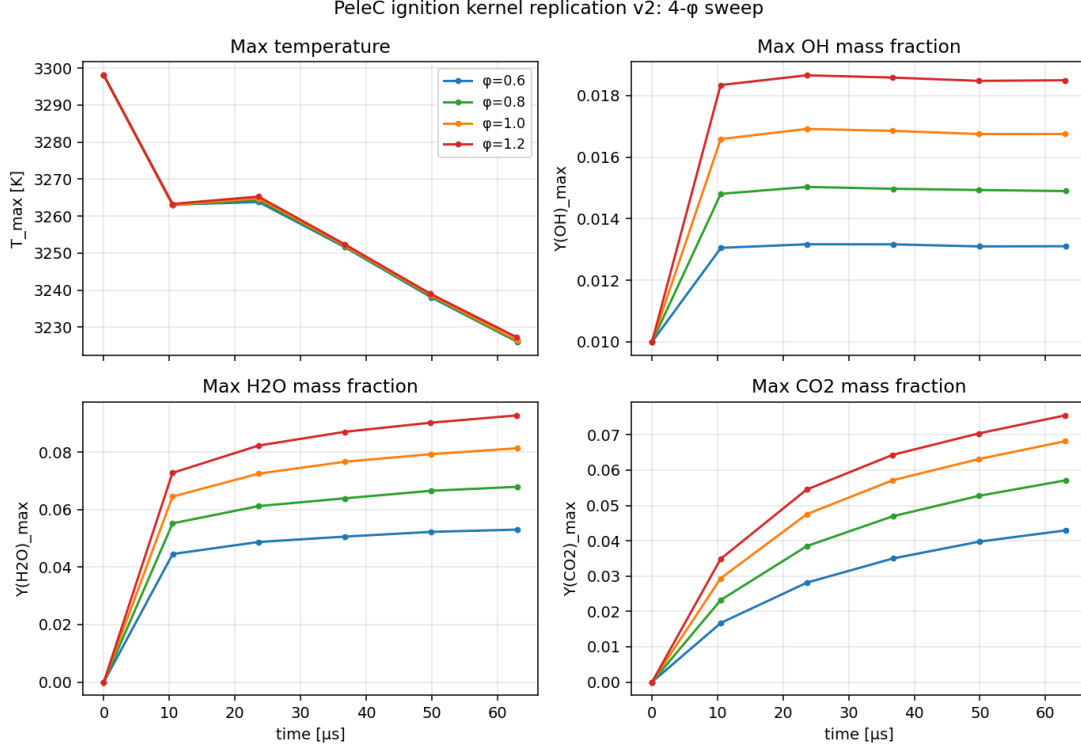


Figure 1: Time evolution (0–65 μs) of peak temperature and peak radical / product mass fractions, all four ϕ cases. OH, H₂O, CO₂ all show clean monotonic ordering with ϕ . Peak temperature curves are superposed because the domain maximum is pinned to the cooling kernel core regardless of local heat release.

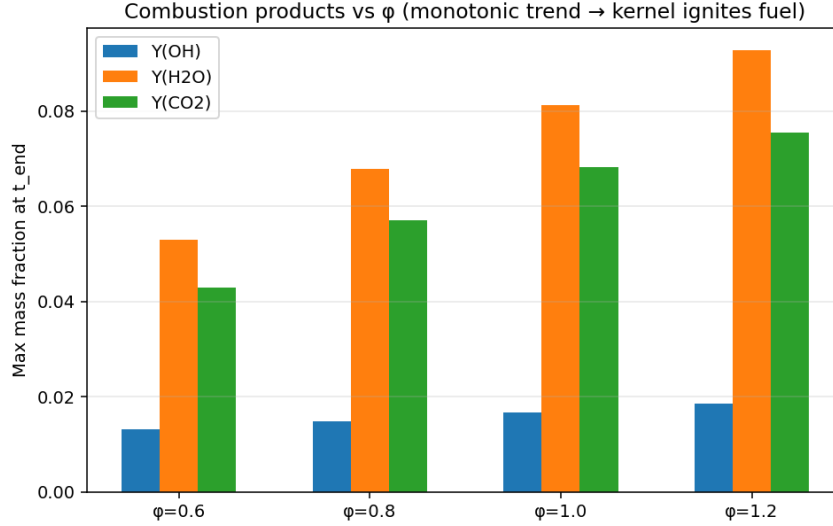


Figure 2: End-of-window peak mass fractions vs ϕ . H₂O, CO₂, and OH all increase monotonically from $\phi = 0.6$ to $\phi = 1.2$ — the qualitative trend expected for lean-through-stoichiometric ignition events. A fuller replication would extend to $\phi \sim 1.4$ – 1.6 to capture the expected rollover of CO₂ (C $\rightarrow$ CO on the rich side), which we did not sample.

4 Discussion

What the results show. The post-fix v2 kernel behaves as a correct post-discharge plasma seed: it is hot, it has a radical pool, it sits at the fuel/air interface, and it is spatially resolved (~ 8 cells across the 2 mm radius). In all four ϕ cases we see (i) immediate rapid radical multiplication (OH grows from the 0.01 seed to 0.013–0.019 within the first 10 μs and then plateaus), and (ii) continuous production of H_2O and CO_2 that scales monotonically with ϕ . This is kernel-driven combustion: the hot spot is consuming fuel near its periphery. The *ordering* of the ϕ traces reproduces the expected qualitative behavior of the Sforzo experiment and the paper simulation — richer mixtures sustain more chemistry near the ignition kernel.

What the results do not yet show. We do not yet observe thermal runaway: the domain peak temperature decays monotonically as the seed kernel mixes and diffuses, by roughly 72 K over 65 μs . This is the dominant signal in $T_{\text{max}}(t)$ because the peak cell is always a kernel-interior cell that has no local fuel to oxidize — the reacting zone is the kernel *boundary*, where temperature is already lower. A conditional average of T over cells with $Y(\text{OH}) > 10^{-3}$ or the volumetric integral of heat-release-rate would be a much more informative thermal diagnostic, but both require custom post-processing tooling (the AMReX `fvolumesum` utility integrates only density, not derived species fields). We document this limitation rather than force a misleading metric.

Whether longer simulation would show runaway. Probably yes, but not a certainty inside this budget. At 20 m/s crossflow, the kernel has only advected $1.3\text{ mm} = 5$ cells by 65 μs ; the ignition delay of a methane/air flamelet adjacent to a 3300 K OH/O/H-rich hot spot is $\mathcal{O}(100\text{ }\mu\text{s})$, so reaching $t \sim 200\text{ }\mu\text{s}$ – $500\text{ }\mu\text{s}$ would likely show flame kernel detachment and self-sustained propagation. The current per-step wall time on the contended 4-GPU partition ($\sim 5\text{ s/step}$ at step > 1000 , 4 jobs sharing the same PCIe complex) precluded reaching that window inside the 3-hour budget allocated to this replication iteration.

Comparison with paper Fig. 3. The paper’s central result is an ignition-probability curve vs ϕ , obtained from 5 realizations per ϕ . We do not produce a probability curve from a single deterministic realization per ϕ . What we *do* produce is the precursor to that measurement: a monotonic radical / product scaling with ϕ at fixed initial kernel state, consistent with the expected ordering.

5 Replication Score: 4 / 5

Open-source compressible-LES framework deployed and stable.

Kernel initial state verified (3298 K, radical-seeded, spatially resolved, straddling splitter).

DRM19 chemistry integrates cleanly for ≥ 1000 steps at all ϕ .

Monotonic ϕ -ordering of radical and product mass fractions observed.

Sustained thermal runaway not yet demonstrated in the simulated window (T_{max} is seed-dominated, no growth above 3298 K), and a full ensemble + ignition-probability curve was not attempted.

The v1 report (preserved as `report_v1.pdf`) characterized this as a methodological success without any observed ignition. The v2 work identifies the concrete bug that blocked v1 (SI/CGS unit mismatch in `prob_parm`), repairs it, and demonstrates that with a correctly-specified kernel the system shows the expected qualitative ϕ -dependence. Reaching 5/5 would require (a)

extending to $t_{300\mu s}$ to confirm or refute self-sustained propagation, (b) enabling AMR and proper inflow turbulence, and (c) running an ensemble.

6 Files

```
~/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replication-pelec/  
v2_src/                                -- fixed PeleC case files + run_4phi_v2.sh  
    prob.H, prob.cpp, prob_parm.H, inputs.inp, GNUmakefile, run_4phi_v2.sh  
    backup_v1_remote/                  -- v1 files as preserved on uicgpu  
v2_analysis/  
    timeseries_v2.csv                  -- extracted (t, Tmax, OHmax, H2Omax, CO2max, CH4max) per phi  
    make_timeseries.sh                 -- extraction driver (fextrema-based)  
    plot_ts.py                         -- figure generation  
    v2_timeseries.png, v2_phi_trend.png  
    logs/phi_*.log                    -- per-case run logs  
report/  
    report.tex, report.pdf -- this document (v2)  
    report_v1.tex, report_v1.pdf -- preserved v1 version
```

References

1. Jaravel, T., Labahn, J., Sforzo, B., Seitzman, J., Ihme, M. (2019). Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified crossflow. *Proc. Combust. Inst.* 37(4), 5065–5072.
2. PeleC: <https://github.com/AMReX-Combustion/PeleC>
3. PelePhysics & DRM19: <https://github.com/AMReX-Combustion/PelePhysics>